MeaProcess-Runtime Scripting API

In the scheme editor a scripting Input Block and a scripting Output Block are available. With the scripting Input Block signal values can be set from a script and with the scripting Output Block signal values can be read. The scripting Input Block must have signals defined into the output port. The signals can be defined through a double click of the scripting Input Block output port. After the scripting blocks are defined than the scheme can be used into a script. Following scripts are supported: VBScript, Java Script, Phyton, Perl, PHP.

MeaProcessApp.MeaProcess Object

For using this object into a Programming language following TLB file is into the Runtime installation folder available: “Mp.Runtime.Scripting.tlb”. The API of this object is:

```
bool OpenScheme(string scheme, int port);

Start the MeaProcess runtime and opens a scheme file.

Parameter:

scheme        The scheme path and file name (c:\\test\\demo.mpw)

port          The communication port between the script and MeaProcess runtime, default
              5050

Return values:

True if the runtime was started successful
```

```
void Start();

Start the execution of the scheme.
```

```
void Stop();

Stop the execution of the scheme.
```

```
void Reinitialize();

Reinitialize the scheme.
```

```
int InputSignals

This property returns the number of the input signals available into the Scripting
Input Block.
```

```
int OutputSignals
```

This property returns the number of the output signals available into the Scripting output block.

```
bool Visible
```

This property sets the runtime MainFrame visible.

```
string Messages;
```

This property is read only. Read the runtime messages. The string has the following format:

```
MsgType\t[TimeStamp] Message\n
```

Where MsgType = ERROR, WARNING, INFO and
TimeStamp = System formatted Date and Time

```
ISignal GetSignal(int index, bool input);
```

This function returns a signal by the index.

Parameter:

index	The signal index.
input	True if a input signal is requested. False for an output signal.

Return:

A signal object.

```
void Close();
```

Close the MeaProcess runtime.

MeaProcessApp.Signal Object

With this object signal properties can be read or set. The API of this object is:

```
string Name
```

This is a read only property. Read the signal name.

```
string Unit
```

This is a read only property. Read the signal unit.

```
double Value
```

Read or write the signal value.

```
double Minimum
```

This is a read only property. Read the signal physical minimum.

```
double Maximum
```

This is a read only property. Read the signal physical maximum.

Visual Basic Script Example

```
'For x64 execute with "C:\Windows\Syswow64\wscript.exe"

'Create a MeaProcess Runtime object
Set obMea =  WScript.CreateObject("MeaProcessApp.MeaProcess")

'Open a scheme, give the runtime a port for communication with this script object
obMea.OpenScheme "C:\\Users\\cr\\Desktop\\test.mpw", 5050

'Start the measurement
obMea.Start

'Read some signal data out
for i = 0 to obMea.OutputSignals -1
    Set signal = obMea.GetSignal(i, false)

    for j = 0 to 2
        WScript.echo signal.Value
        WScript.sleep 1000
    next
next


'Write a signal into the system
for j = 0 to 10
       for i = 0 to obMea.InputSignals -1
               Set signal = obMea.GetSignal(i, true)
               signal.Value = 1
               WScript.sleep 500
               signal.Value = 0
               WScript.sleep 500
       next
next

'Stop the measurement
obMea.Stop

'Close the runtime
obMea.Close
```